

Theoretical Developments in Gravitational Wave Physics, Kyoto '26

KERR GEOMETRY FROM WORLDLINES, HIGHER-SPIN QFT AND SCATTERING AMPLITUDES

[arXiv:2212.06120, 2312.14913]

[arXiv:2410.23271, 2512.24549]

[arXiv:2605.29831]

M. Ben-Shahar, L. Bohnenblust, M. Chiodaroli, H. Johansson, P. Pichini, A. Ochirov, E. Skvortsov, I. Surubaru

Lucile Cangemi
University of Edinburgh



THE UNIVERSITY
of EDINBURGH

CONTENTS

- ❖ Set-up + refresh non-spinning approaches
- ❖ EFTs for general spinning compact bodies
- ❖ Kerr EFTs @ linear order
- ❖ Kerr EFTs @ higher orders

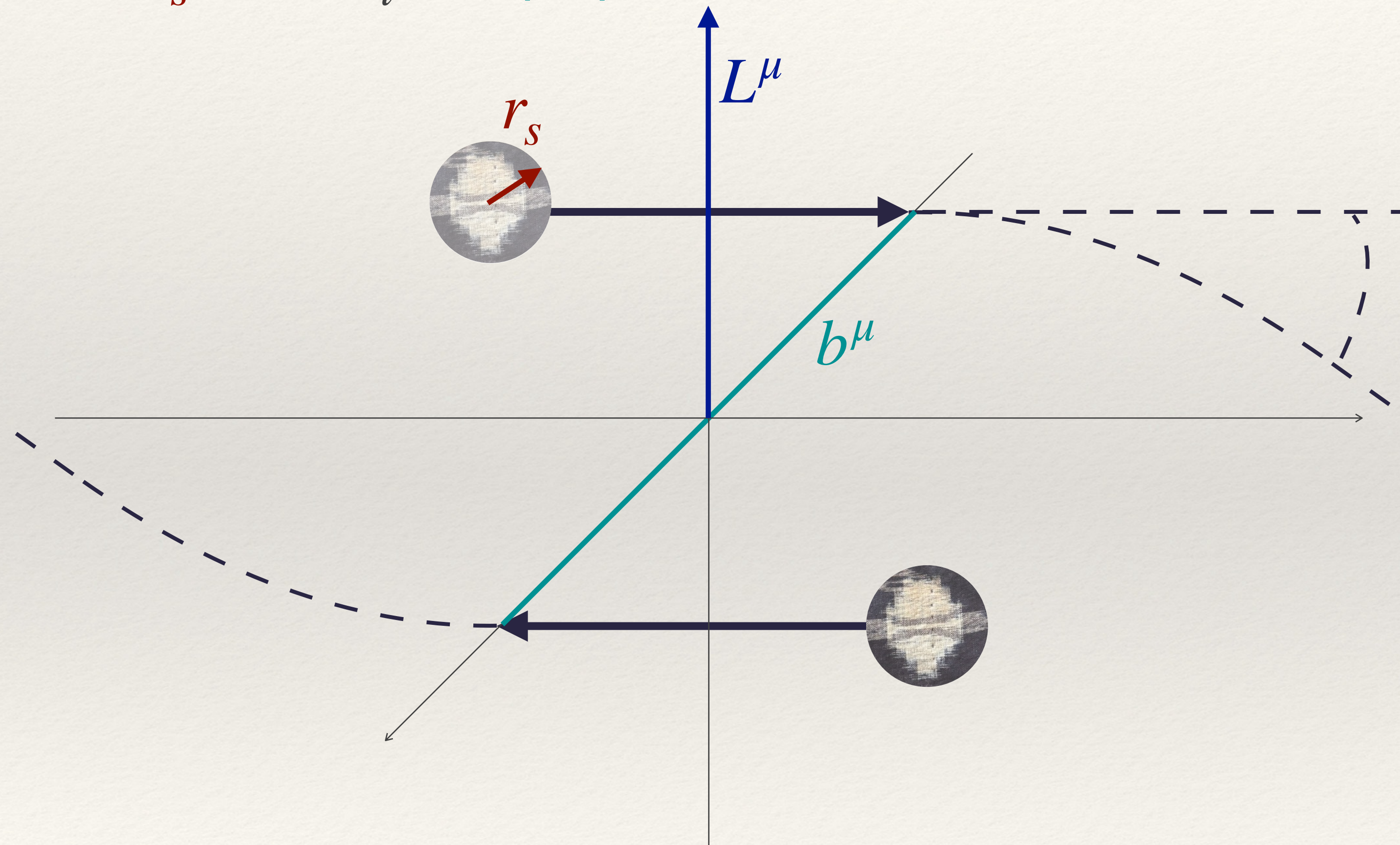
CONTENTS

- ❖ Set-up + refresh non-spinning approaches
- ❖ EFTs for general spinning compact bodies worldline, QFTs, scattering amps
- ❖ Kerr EFTs @ linear order mass + current multipoles, matching vs first principles
- ❖ Kerr EFTs @ higher orders Teukolsky solutions, higher-spin proposals & 2PM observables

SET UP

classical weak-field

GR $r_s = Gm_i \ll |b|$



Newtonian gravity

$$\theta = \theta^{(1)}G + \theta^{(2)}G^2 + \mathcal{O}(G^3)$$

GR corrections $\sim Gm|b|$

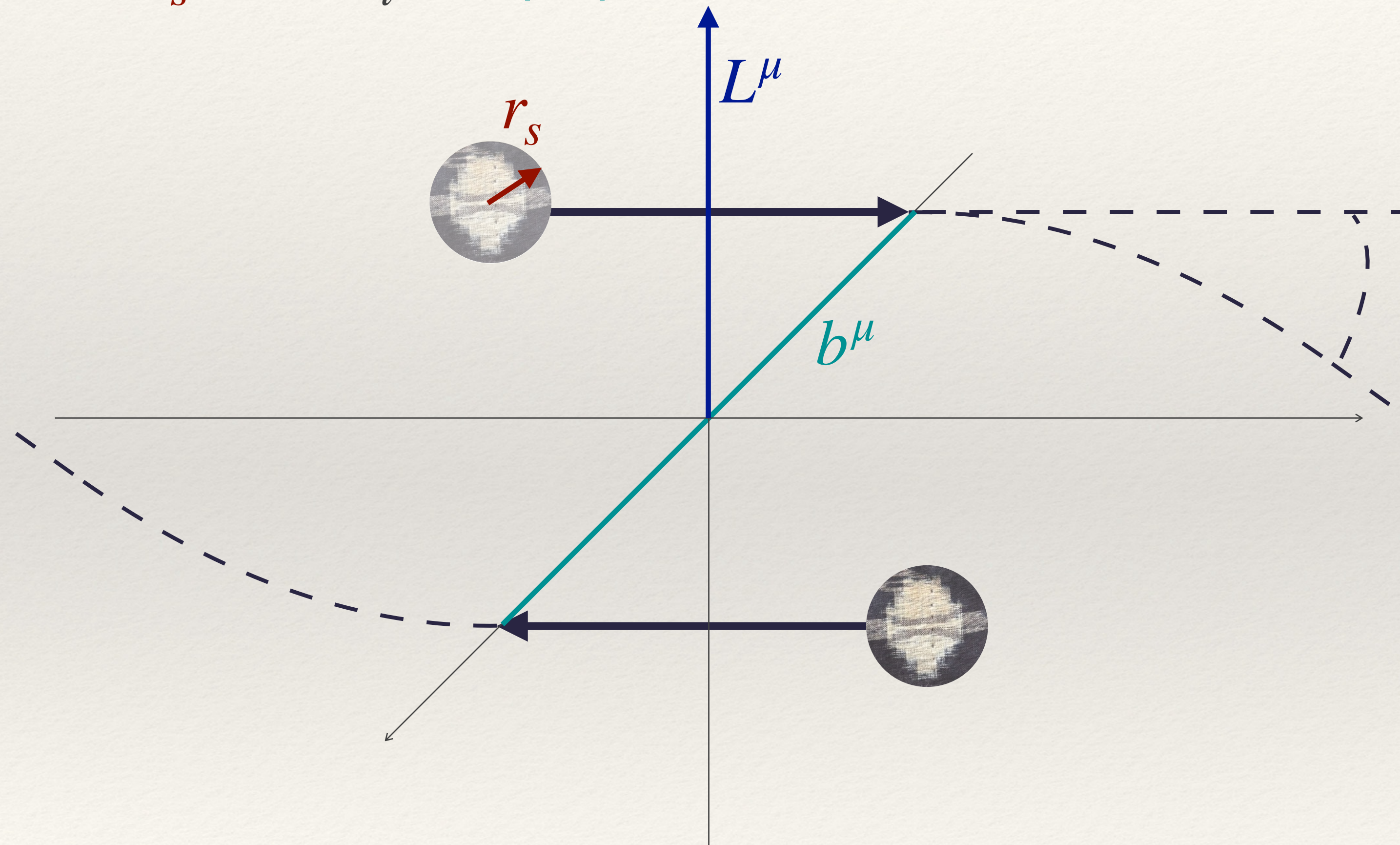
post-Minkowski expansion

see talks by Thibault, Fei, Donal...

SET UP

classical weak-field

GR $r_s = Gm_i \ll |b|$ ← well-defined asymptotic states!



Newtonian gravity

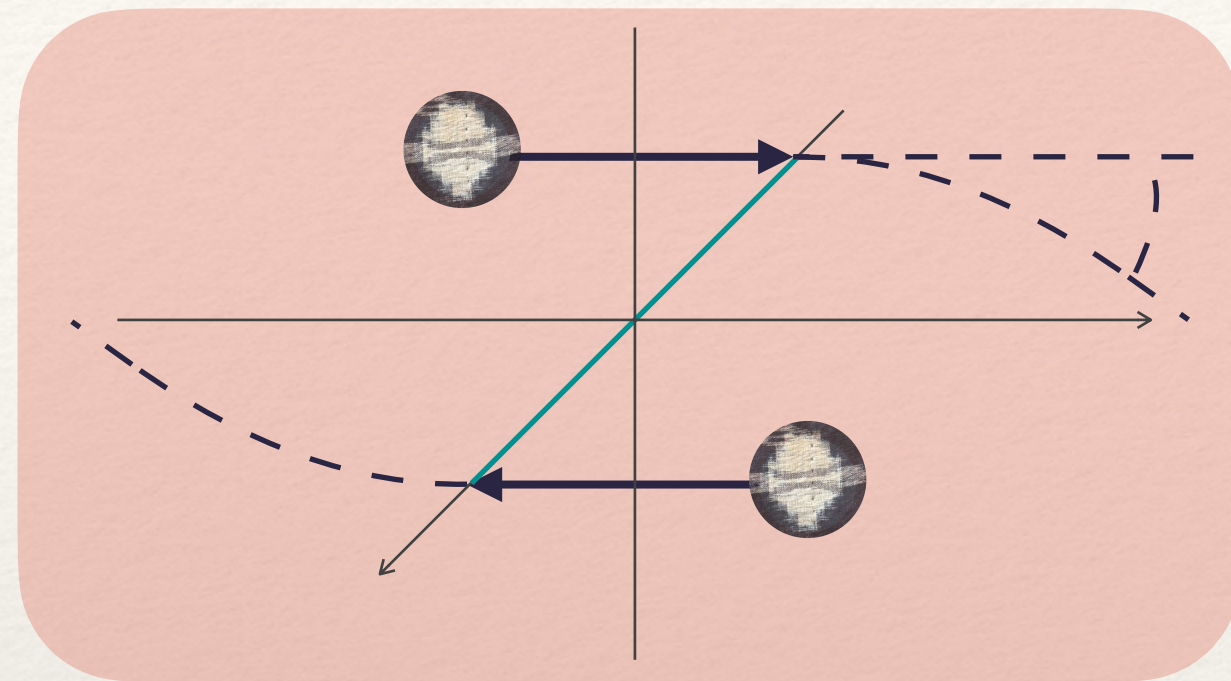
$$\theta = \theta^{(1)}G + \theta^{(2)}G^2 + \mathcal{O}(G^3)$$

GR corrections $\sim Gm|b|$

post-Minkowski expansion

see talks by Thibault, Fei, Donal...

POINT-PARTICLE EFT



$$S_{EFT} = \sum_i S_{pp,i} + \int d^4x \sqrt{-g} R$$

Worldline

$$S_{pp} = - \int d\tau \, p_{i,\mu} \dot{x}_i^\mu - \ell(p_i^2 - m_i^2)$$

captures dynamics of scattering of
Schwarzschild BHs!

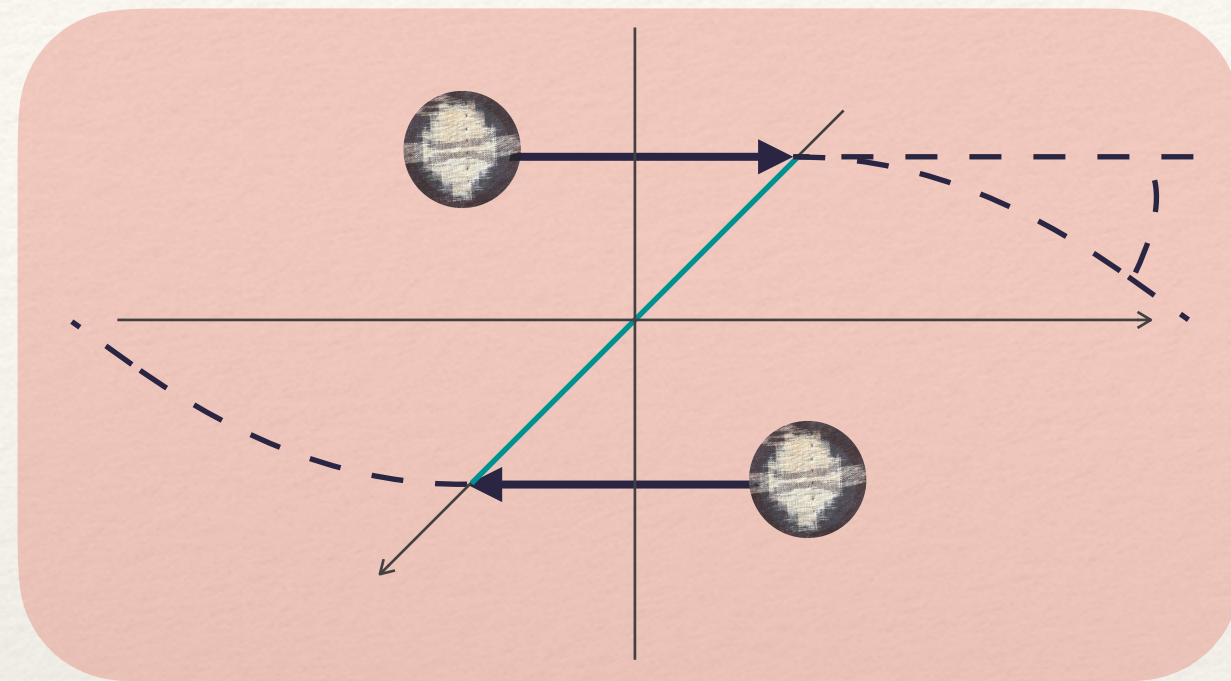
finite size effects correct $S_{p.p.}$ @ $\mathcal{O}(G^7)$

Field theory

$$S_{pp} = \int d^4x \, \nabla^\mu \phi_i \nabla_\mu \phi_i - m_i^2 \phi^2$$

see talks by Thibault, Fei, Donal...

POINT-PARTICLE EFT



$$S_{EFT} = \sum_i S_{pp,i} + \int d^4x \sqrt{-g} R$$

Worldline

$$S_{pp} = - \int d\tau \, p_{i,\mu} \dot{x}_i^\mu - \ell(p_i^2 - m_i^2)$$

Field theory

$$S_{pp} = \int d^4x \, \nabla^\mu \phi_i \nabla_\mu \phi_i - m_i^2 \phi^2$$

can compute in PM expansion:

- ❖ conservative Hamiltonian
- ❖ radial action / eikonal phase
- ❖ observables: θ , Δp_μ

$\exists$ diagrammatic organisation

see talks by Thibault, Fei, Donal...

POINT-PARTICLE EFT

Example: impulse $\Delta p^\mu = \langle \psi_{out} | \hat{\mathbb{P}}^\mu | \psi_{out} \rangle$

KMOC: in-in expectation values of
observables with suitable wave fns.

for non-spinning

$$= p_{CM}^\mu (\cos \theta - 1)$$

see talk by Donal, Hyon...

POINT-PARTICLE EFT

Example: impulse $\Delta p^\mu = \langle \psi_{out} | \hat{\mathbb{P}}^\mu | \psi_{out} \rangle$

KMOC: in-in expectation values of observables with suitable wave fns.

diagrammatic expansion:

see talk by Donal, Hyon...

@ 1PM $i \int d^D \mu e^{iq \cdot b} q^\mu \left[\begin{array}{ccc} \bar{p}_1 & \text{---} \bullet & \bar{p}_1 + q \\ & \text{---} \text{wavy} & \\ \bar{p}_2 & \text{---} \bullet & \bar{p}_2 + q \end{array} \right]$

@ 2PM $i \int d^D \mu e^{iq \cdot b} \left[q^\mu \begin{array}{c} \bullet \text{---} \bullet \\ \text{wavy} \text{---} \bullet \\ \bullet \end{array} - \int d\Phi(l_1) d\Phi(l_2) (l_1 - \bar{p}_1)^\mu \begin{array}{c} \bullet \text{---} \bullet \\ \text{wavy} \text{---} \bullet \\ \bullet \end{array} \right]$

POINT-PARTICLE EFT

Example: impulse $\Delta p^\mu = \langle \psi_{out} | \hat{\mathbb{P}}^\mu | \psi_{out} \rangle$

KMOC: in-in expectation values of observables with suitable wave fns.

diagrammatic expansion:

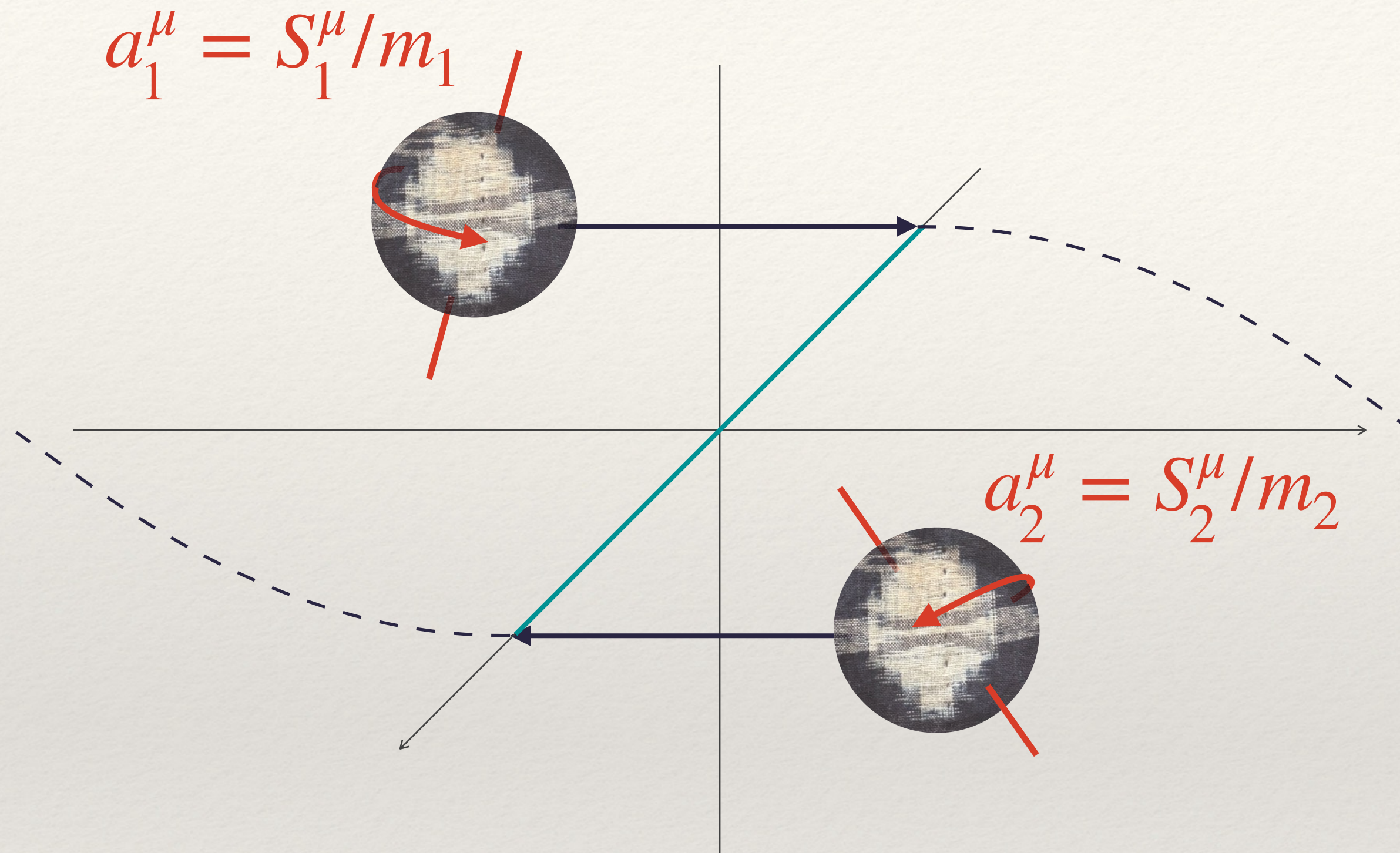
see talk by Donal, Hyon...

@ 1PM $i \int d^D \mu e^{iq \cdot b} q^\mu \left[\begin{array}{c} \bar{p}_1 \text{---} \bullet \text{---} \bar{p}_1 + q \\ \text{wavy line} \\ \bar{p}_2 \text{---} \bullet \text{---} \bar{p}_2 + q \end{array} \right]$

@ 2PM $i \int d^D \mu e^{iq \cdot b} \left[q^\mu \begin{array}{c} \bullet \text{---} \bullet \\ \text{wavy line} \\ \bullet \end{array} - \int d\Phi(l_1) d\Phi(l_2) (l_1 - \bar{p}_1)^\mu \begin{array}{c} \bullet \text{---} \bullet \\ \text{wavy line} \\ \bullet \end{array} \right]$

n-PM impulse — need coupling of n - gravitons to massive particle

ADDING SPIN



dynamics not generically in a plane!

$$\Delta p_{\parallel}^{\mu} = p_{CM}^{\mu} (\cos \theta - 1)$$

$$\Delta p_{\perp}^{\mu} = |p_{CM}| \sin \theta (\cos \phi \hat{b}^{\mu} + \sin \phi \hat{L}^{\mu})$$

new scale $a_i \Rightarrow$ spin multipolar expansion

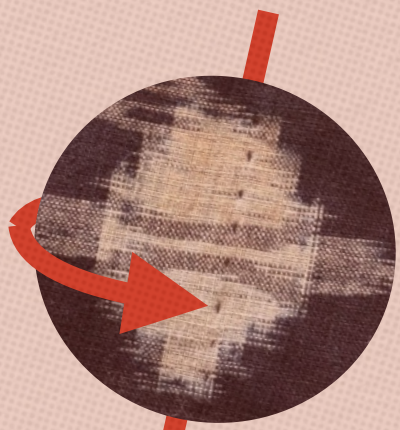
$$\mathcal{O}^{(n)} = \mathcal{O}^{(n,0)} + \mathcal{O}_{\mu_1}^{(n,1)} a_1^{\mu_1} + \mathcal{O}_{\mu_1 \mu_2}^{(n,2)} a_1^{\mu_1} a_1^{\mu_2} + \dots$$

naive PM counting



! physical BHs satisfy $a_i \leq Gm_i$

ADDING SPIN



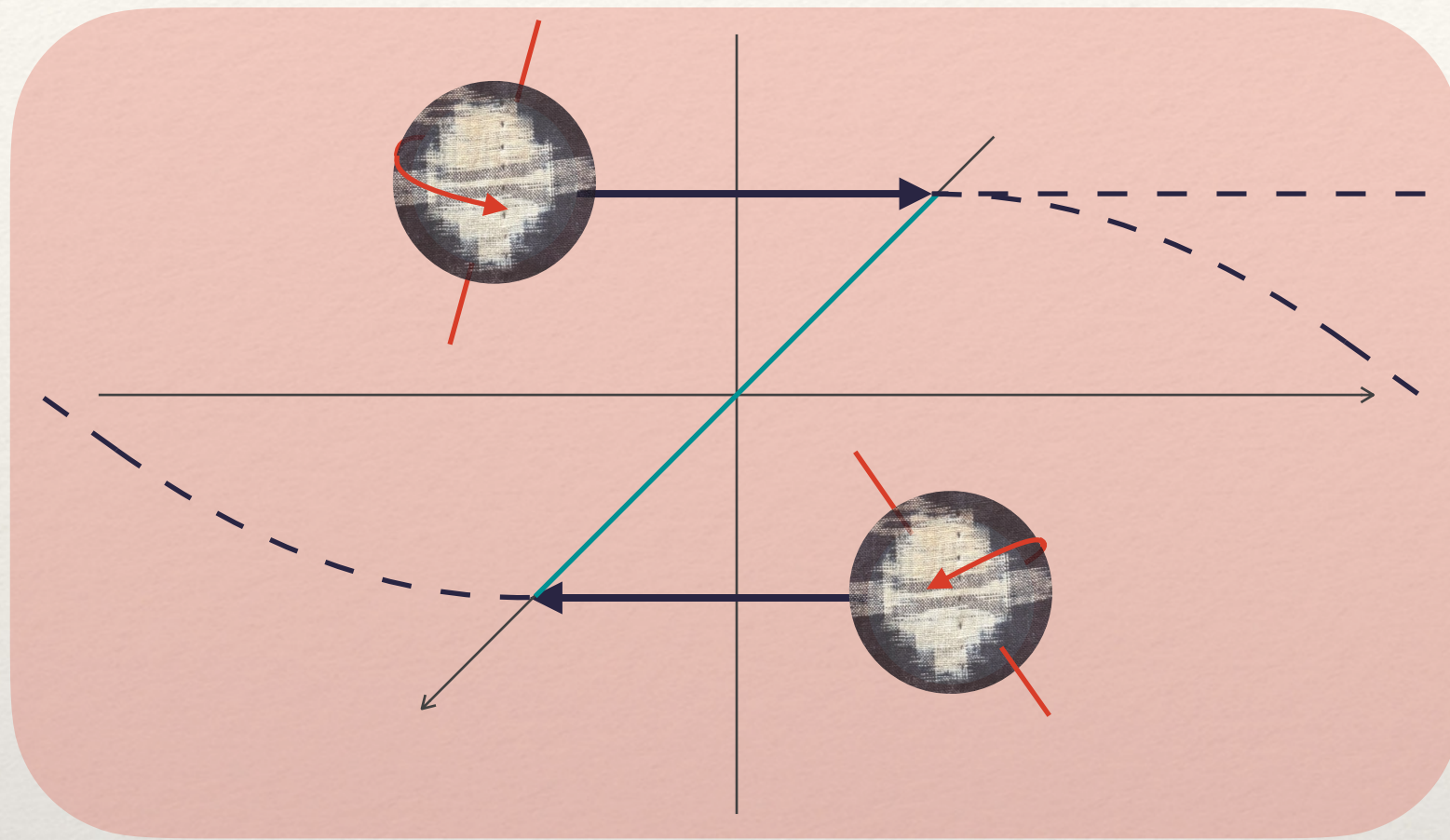
$a_1^\mu =$

	S ⁰ (Spin-0)	S ¹ (Spin-1/2)	S ² (Spin-1)	S ³ (Spin-3/2)	S ⁴ (Spin-2)	S ⁵ (Spin-5/2)
1PM (tree level)	G	G ²	G ³	G ⁴	G ⁵	G ⁶
2PM (1 loop)	G ²	G ³	G ⁴	G ⁵	Guevara, Ochirov, Vines '18 Chen, Chung,Huang, Kim '21 Haddad, Jakobsen, GM, Plefka '24	Bern, Kosmopoulos, Luna, Roiban, Teng '22; Aoude, Haddad, Helset '23; Bautista '23; Chen, Wang '24; Bohnenblust, Cangemi, Johansson, Pichini '24
3PM (2 loops)	Bern, Cheung, Parra-Martinez, Ruf, Herrmann, Roiban, Shen, Solon, Zeng, Kälin, Liu, Porto, Di Vecchia, Heissenberg, Russo, Veneziano, Travaglini, Brandhuber, Damgaard, Planté, Vanhove, Bjerrum-Bohr	Jakobsen, GM '22 Akpinar, Cordero, Kraus, Ruf, Zeng '24	Jakobsen, GM '22 Akpinar, Cordero, Kraus, Ruf, Zeng '24	Akpinar, Cordero, Kraus, Smirnov, Zeng '25	Akpinar, Cordero, Kraus, Smirnov, Zeng '25	G ⁸
4PM (3 loops)	Dlapa, Kälin, Liu, Neef, Porto, Damgaard, Hansen, Planté, Vanhove, Bern, Parra-Martinez, Roiban, Ruf, Shen Solon, Zeng	Jakobsen, GM, Plefka, Sauer, Xu '23	G ⁶	G ⁷	G ⁸	Tail effect log v's
5PM (4 loops)	Driesse, Jakobsen, Klemm, GM, Nega, Plefka, Sauer, Usovitsch '24	G ⁶	G ⁷	G ⁸	G ⁹	G ¹⁰

physical BHs satisfy $a_i \leq Gm_i$

Table from Mogull's slides 2nd
Annual Workshop on Self-force
and amplitudes

ADDING SPIN



$$\longrightarrow S_{EFT} = \sum_i S_{Kerr,i}^? + \int d^4x \sqrt{-g} R$$

ADDING SPIN



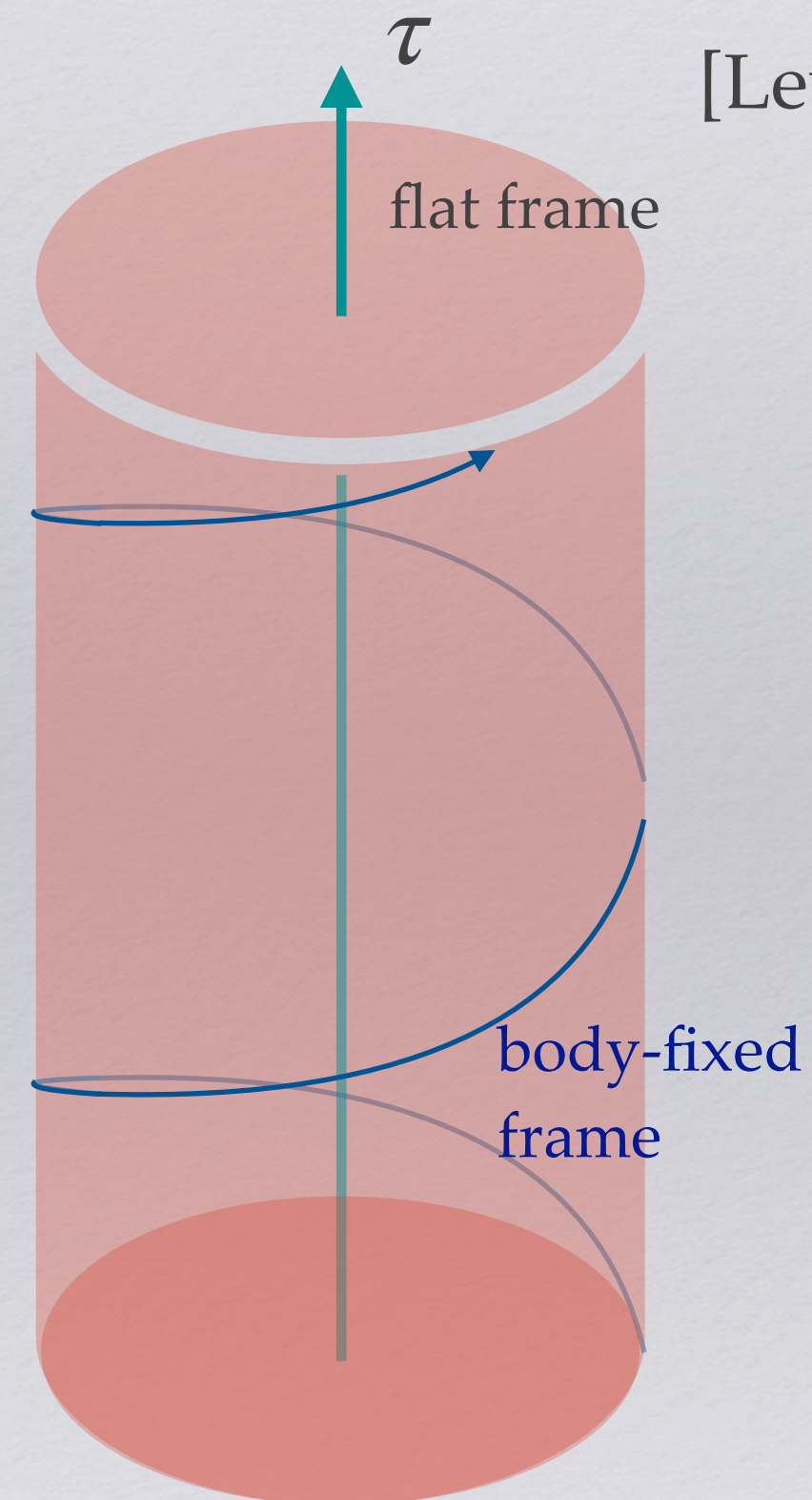
$S_{spin,i}$ general compact spinning bodies (m_i, a_i^μ)

Worldline EFTs

spin tensor S_{ab}

frame fields Λ_I^a

$$e_\mu^I = e_a^\mu \Lambda_I^a$$



[Levi, Steinhoff '15]

ADDING SPIN



$S_{spin,i}$ general compact spinning bodies (m_i, a_i^μ)

Worldline EFTs

spin tensor S_{ab} 6 d.o.f.

constraint! (SSC)

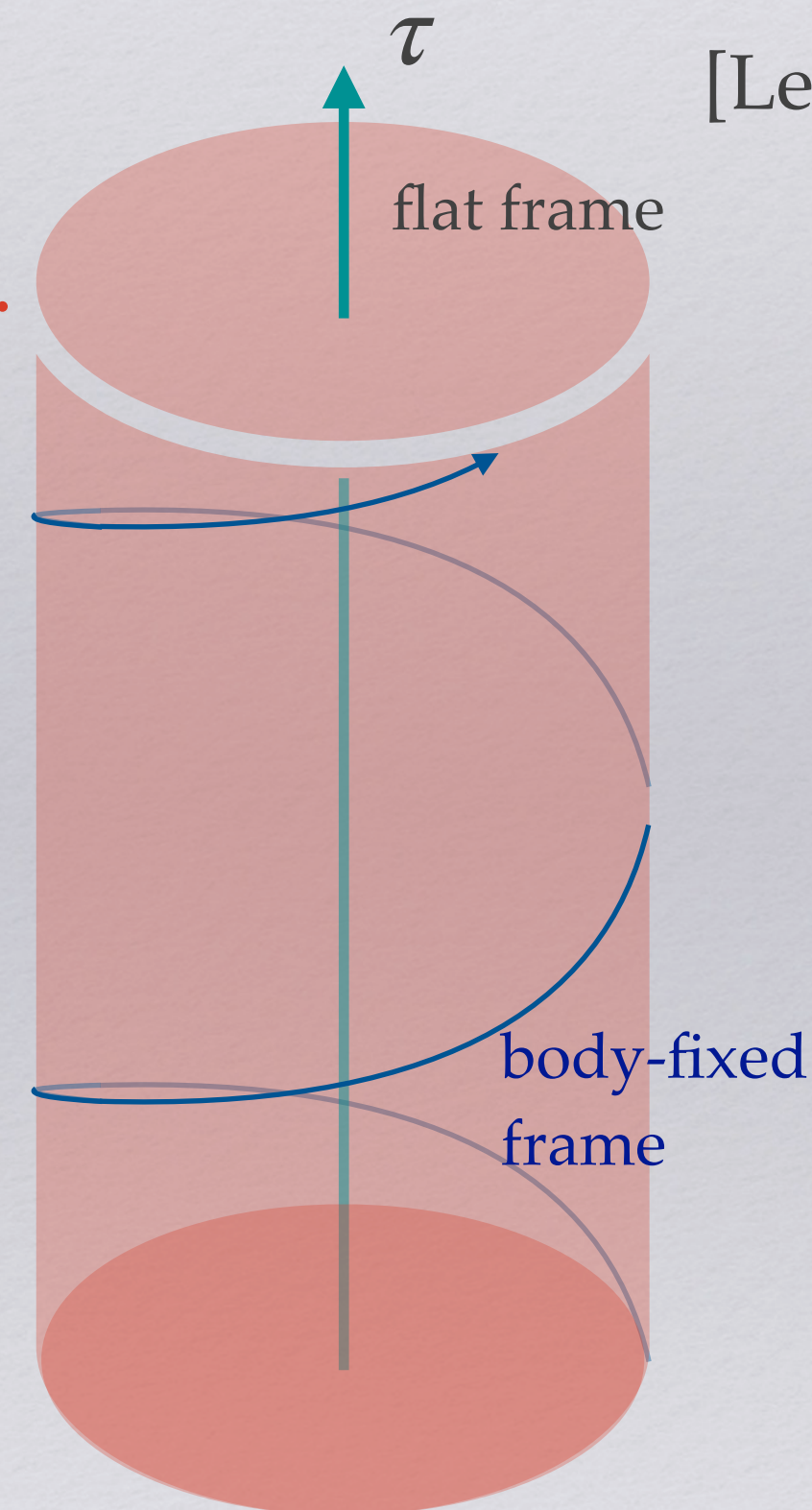
$$p^a S_{ab} = 0$$

spin vector

$$S^a = \epsilon^{abcd} p_b S_{cd}$$

frame fields Λ_I^a

$$e_\mu^I = e_a^\mu \Lambda_I^a$$



[Levi, Steinhoff '15]

kinetic terms

$$S_{pp} + \int d\tau \frac{1}{2} S_{ab} \Omega^{ab} - \frac{1}{|p|} \frac{Dp^\mu}{d\tau} S_{\mu\nu} \Lambda_0^\nu$$

ang. velocity

$$\Omega^{ab} = \Lambda^{aI} \frac{d\Lambda_I^b}{d\tau}$$

interactions $\mathcal{L}_{R^n}(p_\mu, S^\mu, \nabla_\mu, E_{\mu\nu}, B_{\mu\nu})$

D=4: two indep. projections of R

$$E_{\mu\nu} = R_{\mu p \nu p} \quad B_{\mu\nu} = \frac{1}{2} \epsilon_{\nu p}{}^{\kappa\lambda} R_{\mu p \kappa\lambda}$$

ADDING SPIN



$S_{spin,i}$ general compact spinning bodies (m_i, a_i^μ)

Worldline EFTs

spin tensor S_{ab}

constraint! (SSC)

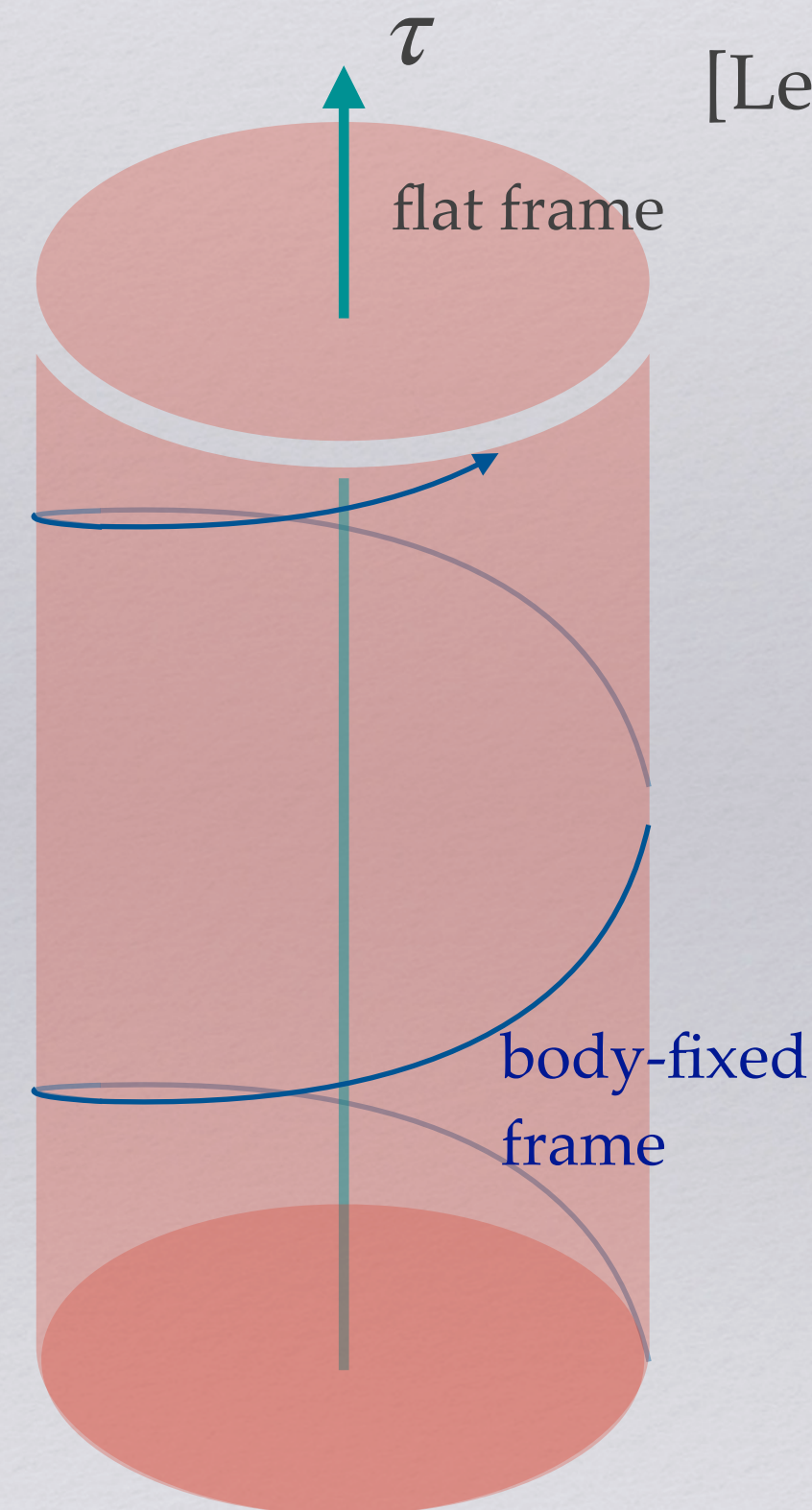
$$p^a S_{ab} = 0$$

spin vector

$$S^a = \epsilon^{abcd} p_b S_{cd}$$

frame fields Λ_I^a

$$e_\mu^I = e_a^\mu \Lambda_I^a$$



[Levi, Steinhoff '15]

kinetic terms

$$S_{pp} + \int d\tau \frac{1}{2} S_{ab} \Omega^{ab}$$

interactions $\mathcal{L}_{R^n}(p_\mu, S^\mu, \nabla_\mu, E_{\mu\nu}, B_{\mu\nu})$

$$\mathcal{L}_R \sim \sum_{j=1}^{\infty} \frac{\boxed{c_{ES^{2j}}}}{(2j)!} (\nabla_\mu)^{2j-2} E_{SS} (S^\mu)^{2j-2} + \sum_{j=1}^{\infty} \frac{\boxed{c_{BS^{2j+1}}}}{(2j+1)!} (\nabla_\mu)^{2j-1} B_{SS} (S^\mu)^{2j-1}$$

Wilson coeffs.

ADDING SPIN



$S_{spin,i}$ general compact spinning bodies (m_i, a_i^μ)

spinning QFTs $\phi \rightarrow \phi^{(s)}$ spin- s fields

❖ spinning EFT

see talk by Fei

[Bern, Luna, Roiban, Shen, Zeng, '20]

[Alaverdian, Bern, Kosmopoulos, Luna, Roiban, Scheopner, Teng '25]

on-shell d.o.f: $\mathcal{E}(p) \sim \varepsilon^{(s)} + \varepsilon^{(s-1)} + \dots + \varepsilon^{(1)}$

❖ fixed spin- s QFTs

massive Weyl spinor $u^a(p) = \begin{pmatrix} |p^a\rangle \\ |p^a] \end{pmatrix}$

on-shell d.o.f: $\varepsilon_p^{(s)} \sim (\varepsilon_p^{(1)})^s \sim (|p\rangle)^{2s}$

$$(p \cdot \sigma) |p^a] = m |p^a\rangle$$

ADDING SPIN



$S_{spin,i}$ general compact spinning bodies (m_i, a_i^μ)

spinning QFTs $\phi \rightarrow \phi^{(s)}$ spin- s fields

❖ spinning EFT **see talk by Fei**

[Bern, Luna, Roiban, Shen, Zeng, '20]

[Alaverdian, Bern, Kosmopoulos, Luna, Roiban, Scheopner, Teng '25]

kinetic term $\phi^{(s)}(\nabla^2 - m^2)\phi^{(s)}$

interactions $H R_{abcd} \phi^{(s)} M^{ab} M^{cd} \phi^{(s)} + \dots$

cl. spin: $\varepsilon_1 \cdot \{M^{\mu_1\nu_1}, \dots, M^{\mu_n\nu_n}\} \cdot \varepsilon_2 \sim S^{\mu_1\nu_1} \dots S^{\mu_n\nu_n} + \mathcal{O}(q^{1-n})$

ADDING SPIN



$S_{spin,i}$ general compact spinning bodies (m_i, a_i^μ)

spinning QFTs $\phi \rightarrow \phi^{(s)}$ spin- s fields

❖ fixed spin- s QFTs [Arkani-Hamed, Huang, Huang '17][Guevara, Ochirov, Vines '18] [Chung, Huang, Kim, Lee '18]
[Chiodaroli, Johansson, Pichini '21]

minimal th. for $s \leq 2$ Dirac, Proca, massive graviton, KK graviton

+ interactions $\mathcal{L}_{R^n}$

ADDING SPIN



$S_{spin,i}$ general compact spinning bodies (m_i, a_i^μ)

spinning QFTs $\phi \rightarrow \phi^{(s)}$ spin- s fields

❖ fixed spin- s QFTs [Arkani-Hamed, Huang, Huang '17][Guevara, Ochirov, Vines '18] [Chung, Huang, Kim, Lee '18]
[Chiodaroli, Johansson, Pichini '21]

minimal th. for $s \leq 2$ Dirac, Proca, massive graviton, KK graviton

+ interactions $\mathcal{L}_{R^n}$

indep. $\mathcal{L}_R$ operators $\longleftrightarrow$ indep. 3pt scattering amps.

$$\mathcal{M}(1^{(s)}, 2^{(s)}, 3^{(h)})$$

ADDING SPIN



$S_{spin,i}$ general compact spinning bodies (m_i, a_i^μ)

spinning QFTs $\phi \rightarrow \phi^{(s)}$ spin- s fields

[Arkani-Hamed, Huang, Huang '17]

❖ fixed spin- s QFTs $\longleftrightarrow$ spin- s spinor hel. amps.

massive Weyl spinors

$$\mathcal{M}(1^{(s)}, 2^{(s)}, 3^{(-2)}) = (p_1 \cdot \varepsilon_3^{-1})^2 \sum_{n=0}^{2s} g_{n,s} \langle \mathbf{12} \rangle (x^{-1} \langle \mathbf{13} \rangle \langle \mathbf{32} \rangle)^{2s-n}$$

graviton pol.
 $x \sim p_1 \cdot \varepsilon_3$

couplings in $\mathcal{L}_R$

mass + dipole constraints $\implies 2s - 1$ d.o.f = # of Wilson coeffs $\mathcal{O}(S^{n \leq 2s})$

ADDING SPIN



$S_{spin,i}$ general compact spinning bodies (m_i, a_i^μ)

spinning QFTs $\phi \rightarrow \phi^{(s)}$ spin-s fields

❖ fixed spin-s QFTs $\longleftrightarrow$ spin-s spinor hel. amps. $\longleftrightarrow$ spin-operator basis

$$\mathcal{M}(1^{(s)}, 2^{(s)}, 3^{(-2)}) = (p_1 \cdot \epsilon_3^{-1})^2 \sum_{n=0}^{2s} \tilde{g}_{n,s} \langle (k_3 \cdot \hat{a})^{2s} \rangle$$

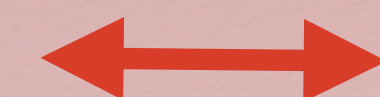
spin-operator: $\hat{a}^{\mu a}_b = m^{-1} (\langle 1^a |)^{2s} \cdot \mathbb{S}^\mu \cdot (| 1_b \rangle)^{2s}$

Pauli-Lubanski op.

$$\mathbb{S}^\mu = \epsilon^\mu_{\nu\rho\sigma} p^\nu \mathbb{M}^{\rho\sigma}$$

spin variables $\longleftrightarrow$ expectations of spin op.

$$g_{n,s}$$



$$\tilde{g}_{n,s}$$

ADDING SPIN



$S_{spin,i}$ general compact spinning bodies (m_i, a_i^μ)

spinning QFTs $\phi \rightarrow \phi^{(s)}$ spin- s fields

❖ fixed spin- s QFTs $\longleftrightarrow$ spin- s spinor hel. amps. $\longleftrightarrow$ spin-operator basis

$$\mathcal{M}_{cl.}(1^{(s)}, 2^{(s)}, 3^{(-2)}) = (p_1 \cdot \varepsilon_3^{-1})^2 \sum_{n=0}^{2s} \tilde{g}_{n,s}(k_3 \cdot a)^{2s}$$

map to classical multipoles

$$n! \tilde{g}_{n,s} \sim \begin{cases} C_{ES^{2j}}, n = 2j \\ C_{BS^{2j+1}}, n = 2j + 1 \end{cases}$$

$2s \geq n \geq 2$

ADDING SPIN



$S_{spin,i}$ general compact spinning bodies $(m_i, a_i^\mu) \rightarrow (C_{ES^{2j}}, C_{BS^{2j+1}})$ or $\tilde{g}_{n,s}$

Worldline EFTs

kinetic terms

$$S_{pp} + \int d\tau \frac{1}{2} S_{ab} \Omega^{ab}$$

interactions $\mathcal{L}_R(p_\mu, S^\mu, \nabla_\mu, E_{\mu\nu}, B_{\mu\nu})$

$$\begin{aligned} \mathcal{L}_R \sim & \sum_{j=1}^{\infty} \frac{c_{ES^{2j}}}{(2j)!} (\nabla_\mu)^{2j-2} E_{SS} (S^\mu)^{2j-2} \\ & + \sum_{j=1}^{\infty} \frac{c_{BS^{2j+1}}}{(2j+1)!} (\nabla_\mu)^{2j-1} B_{SS} (S^\mu)^{2j-1} \end{aligned}$$

QFTs $\phi \rightarrow \phi^{(s)}$ spin-s fields

spinning EFT

$$\mathcal{L}_{kin} = \phi^{(s)} (\nabla^2 - m^2) \phi^{(s)}$$

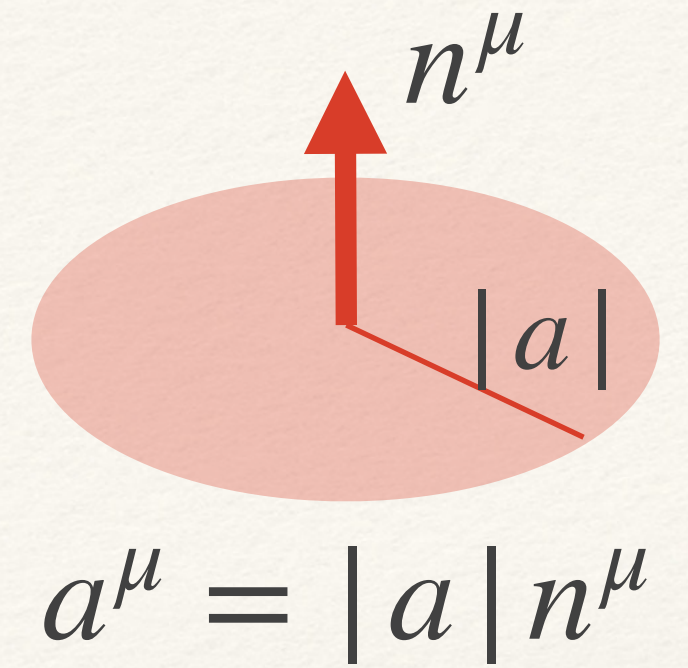
$$\mathcal{L}_R(\phi^{(s)}, M^{\mu\nu}, R)$$

fixed spin-s

$$\mathcal{L}_{kin} \quad s \leq 2 \text{ ✓} \quad s > 2 \text{ ?}$$

$$\mathcal{L}_R \sim \mathcal{M}(1^{(s)}, 2^{(s)}, 3^{(h)}) \rightarrow \tilde{g}_{n,s}$$

SPECIALISING TO KERR



- ❖ linear in R couplings $\longrightarrow$ multipolar expansion of metric $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$

$$h_{\mu\nu}(x) \sim \kappa \int d^d k \frac{e^{-ik \cdot x}}{k^2} T_{\mu\nu}(k)$$

on shell = scattering amp.

$$\varepsilon^\mu \varepsilon^\nu T_{\mu\nu}(k) = (p_1 \cdot \varepsilon)^2 e^{k \cdot a} \quad [\text{Vines '17}]$$

- ❖ matching

Worldline EFTs

$$C_{ES^{2j}} = C_{BS^{2j+1}} = 1$$

QFTs

$$g_{n,s} = \delta_{n,0}$$

$$\mathcal{M}(1^{(s)}, 2^{(s)}, 3^{(-2)}) = (p_1 \cdot \varepsilon_3^{-1})^2 \langle \mathbf{12} \rangle^{2s}$$

WHAT PRINCIPLES FIX KERR EFTs?

* at linear order in R

Wordline EFTs $C_{ES^{2j}} = C_{BS^{2j+1}} = 1$

❖ $\mathcal{N} = 2$ SUSY WL [Jakobsen, Mogull,
Plefka, Steinhoff '21]

→ reproduces $S^{n \leq 2}$

QFTs $g_{n,s} = \delta_{n,0}$

WHAT PRINCIPLES FIX KERR EFTs?

* at linear order in R

Wordline EFTs $C_{ES^{2j}} = C_{BS^{2j+1}} = 1$

❖ $\mathcal{N} = 2$ SUSY WL [Jakobsen, Mogull, Plefka, Steinhoff '21]

→ reproduces $S^{n \leq 2}$

QFTs $g_{n,s} = \delta_{n,0}$

❖ scatt. amps $\mathcal{M}(1^{(s)}, 2^{(s)}, 3^{(-2)})$
 $= (p_1 \cdot \varepsilon_3^{-1})^2 \langle \mathbf{12} \rangle^{2s}$

fixed by best high energy behaviour

[Arkani-Hamed, Huang, Huang '17]

→ reproduces $S^{n \leq 2s}$

[Guevara, Ochirov, Vines '18; Chung, Huang, Kim, Lee '18]

WHAT PRINCIPLES FIX KERR EFTs?

* at linear order in R

Wordline EFTs $C_{ES^{2j}} = C_{BS^{2j+1}} = 1$

❖ $\mathcal{N} = 2$ SUSY WL

→ reproduces $S^{n \leq 2}$

QFTs $g_{n,s} = \delta_{n,0}$

❖ scatt. amps $\mathcal{M}(1^{(s)}, 2^{(s)}, 3^{(-2)})$
 $= (p_1 \cdot \varepsilon_3^{-1})^2 \langle \mathbf{12} \rangle^{2s}$

fixed by best high energy behaviour → reproduces $S^{n \leq 2s}$

❖ minimal spin-s theories $s \leq 2$

→ reproduces $S^{n \leq 4}$

WHAT PRINCIPLES FIX KERR EFTs?

* at linear order in R

Wordline EFTs $C_{ES^{2j}} = C_{BS^{2j+1}} = 1$

❖ $\mathcal{N} = 2$ SUSY WL

→ reproduces $S^{n \leq 2}$

QFTs $g_{n,s} = \delta_{n,0}$

❖ scatt. amps $\mathcal{M}(1^{(s)}, 2^{(s)}, 3^{(-2)})$
 $= (p_1 \cdot \varepsilon_3^{-1})^2 \langle \mathbf{12} \rangle^{2s}$

fixed by best high energy behaviour → reproduces $S^{n \leq 2s}$

❖ minimal spin-s theories $s \leq 2$

→ reproduces $S^{n \leq 4}$

❖ massive higher-spin theories

WHAT PRINCIPLES FIX KERR EFTs?

* at linear order in R

Wordline EFTs $C_{ES^{2j}} = C_{BS^{2j+1}} = 1$

❖ $\mathcal{N} = 2$ SUSY WL

→ reproduces $S^{n \leq 2}$

❖ $\mathcal{N} = 2s$ SUSY WL [Bonocore, Kulesza, Pirsch '25]

→ reproduces $S^{n \leq 2s}$

QFTs $g_{n,s} = \delta_{n,0}$

❖ scatt. amps $\mathcal{M}(1^{(s)}, 2^{(s)}, 3^{(-2)})$
 $= (p_1 \cdot \varepsilon_3^{-1})^2 \langle \mathbf{12} \rangle^{2s}$

fixed by best high energy behaviour → reproduces $S^{n \leq 2s}$

❖ minimal spin-s theories $s \leq 2$

→ reproduces $S^{n \leq 4}$

❖ massive higher-spin theories

HIGHER-SPIN EFTS ^{*} at linear order in R

[LC, Chiodaroli, Johansson, , Ochirov, Pichini, Skvortsov '22, '23]

- ❖ spinning EFT: - have to **match** to get Kerr - **propagates more** than spin-s
- ❖ Goal: build a veritable spin-s QFT, **$s > 2$** + gravity

Non-Chiral Fields $\{ \Phi_{\mu(s)}, \Phi_{\mu(s-1)}, \dots, \Phi_{\mu(0)} \}$

$$+ \delta_0 \Phi_{\mu(k)} = \partial_{(\mu_1} \xi_{\mu_2 \dots \mu_k)} + \mathcal{O}(m) \quad [\text{Zinoviev '01}]$$

Chiral Fields $|\Phi\rangle = \Phi_{\alpha(2s)}$

[Chalmers, Siegel '01; Ochirov, Skvortsov '22]

HIGHER-SPIN EFTS ^{*} at linear order in R

[LC, Chiodaroli, Johansson, , Ochirov, Pichini, Skvortsov '22, '23]

- ❖ spinning EFT: - have to **match** to get Kerr - **propagates more** than spin-s
- ❖ Goal: build a veritable spin-s QFT, **$s > 2$** + gravity

Non-Chiral Fields $\{ \Phi_{\mu(s)}, \Phi_{\mu(s-1)}, \dots, \Phi_{\mu(0)} \}$

$$+ \delta_0 \Phi_{\mu(k)} = \partial_{(\mu_1} \xi_{\mu_2 \dots \mu_k)} + \mathcal{O}(m) \quad [\text{Zinoviev '01}]$$

free theory

$$\mathcal{L}_0 = - \sum_{k=0}^s (-1)^{k+1} \left[\Phi_{\mu(k)} (\square + m^2) \Phi^{\mu(k)} - \frac{k(k-1)}{4} \tilde{\Phi}_{\mu(k-2)} (\square + m^2) \tilde{\Phi}^{\mu(k-2)} \right]$$

$$- \sum_{k=0}^{s-1} (-1)^k (k+1) G_{\mu(k)} G^{\mu(k)}$$

$$G^{\mu(k)} := \partial_\lambda \Phi^{\lambda \mu(k)} - \frac{k}{2} \partial^\mu \tilde{\Phi}^{\mu(k-1)} + m \alpha_k \Phi^{\mu(k)} - m \alpha_{k+1} \frac{k+1}{2} \tilde{\Phi}^{\mu(k)} - m \alpha_k \frac{k-1}{4} \eta^{\mu\mu} \tilde{\Phi}^{\mu(k-2)}$$

Chiral Fields $|\Phi\rangle = \Phi_{\alpha(2s)}$

[Chalmers, Siegel '01; Ochirov, Skvortsov '22]

free theory

$$\mathcal{L}_0 = \langle \partial_\mu \Phi | \partial^\mu \Phi \rangle - m^2 \langle \Phi | \Phi \rangle$$

HIGHER-SPIN EFTS ^{*} at linear order in R

[LC, Chiodaroli, Johansson, , Ochirov, Pichini, Skvortsov '22, '23]

- ❖ spinning EFT: - have to **match** to get Kerr - **propagates more** than spin-s
- ❖ Goal: build a veritable spin-s QFT, $s > 2$ + gravity

Non-Chiral Fields $\{ \Phi_{\mu(s)}, \Phi_{\mu(s-1)}, \dots, \Phi_{\mu(0)} \}$

$$+ \delta_0 \Phi_{\mu(k)} = \partial_{(\mu_1} \xi_{\mu_2 \dots \mu_k)} + \mathcal{O}(m)$$

minimal coupling

breaks massive gauge symm.

$$\delta \mathcal{L} \neq 0$$

Chiral Fields $|\Phi\rangle = \Phi_{\alpha(2s)}$

minimal coupling

breaks parity

HIGHER-SPIN EFTS * at linear order in R

[LC, Chiodaroli, Johansson, , Ochirov, Pichini, Skvortsov '22, '23]

- ❖ spinning EFT: - have to **match** to get Kerr - **propagates more** than spin-s
- ❖ Goal: build a veritable spin-s QFT, $s > 2$ + gravity

Non-Chiral Fields $\{ \Phi_{\mu(s)}, \Phi_{\mu(s-1)}, \dots, \Phi_{\mu(0)} \}$

$$+ \delta_0 \Phi_{\mu(k)} = \partial_{(\mu_1} \xi_{\mu_2 \dots \mu_k)} + \mathcal{O}(m)$$

minimal coupling + $\mathcal{L}_R + \delta_1$

$$\text{constrained by } \delta_0 \mathcal{L}_R + \delta_1 \mathcal{L}_0 = 0$$

power-counting
of operators

Chiral Fields $|\Phi\rangle = \Phi_{\alpha(2s)}$

minimal coupling + $\mathcal{L}_R$

constrained by **parity**

power-counting
of operators

$$\mathcal{L}_R = -\frac{\sqrt{-g}}{4} \sum_k \frac{2s-k-1}{m^{2k}} \langle \Phi [|\overleftarrow{\nabla}| \overrightarrow{\nabla} |^{\odot k} \odot |R_-|] \Phi \rangle$$

! uniquely determine $\mathcal{M}(1^{(s)}, 2^{(s)}, 3^{(-2)}) = (p_1 \cdot \varepsilon_3^{-1})^2 \langle 12 \rangle^{2s}$!

WHAT PRINCIPLES FIX KERR EFTs?

* at linear order in R

Wordline EFTs $C_{ES^{2j}} = C_{BS^{2j+1}} = 1$

❖ $\mathcal{N} = 2$ SUSY WL

→ reproduces $S^{n \leq 2}$

❖ $\mathcal{N} = 2s$ SUSY WL

→ reproduces $S^{n \leq 2s}$

QFTs $g_{n,s} = \delta_{n,0}$

❖ scatt. amps $\mathcal{M}(1^{(s)}, 2^{(s)}, 3^{(-2)})$
 $= (p_1 \cdot \varepsilon_3^{-1})^2 \langle \mathbf{12} \rangle^{2s}$

fixed by best high energy behaviour → reproduces $S^{n \leq 2s}$

❖ minimal spin-s theories $s \leq 2$

→ reproduces $S^{n \leq 4}$

❖ massive higher-spin theories

fixed by consistency + power counting → reproduces $S^{n \leq 2s}$

DIVERSION: HIGHER-DIMENSIONS

- ❖ Why higher-dimensions?  often encounter divergent integrals,
want to use dim. reg. $D = 4 - 2\epsilon$
-  no uniqueness thm: is it reflected in the EFT?

Higher-dim. spinning worldline more types of couplings available

Higher-dim. spinning QFTs more spinning states available

DIVERSION: HIGHER-DIMENSIONS

Higher-dim. spinning worldline

unchanged $\mathcal{L}_{kin'}$ but modified ints:

$$\mathcal{L}_R \sim \sum_{j=1}^{\infty} \frac{c_{1,j}}{(2j)!} (\nabla)^{2j-2} (E_{\mu\nu} (S \cdot S)^{\mu\nu}) (S \cdot S)^{2j-2}$$

spin-tensor

mass multipoles

$$+ \sum_{j=1}^{\infty} \frac{c_{2,j}}{(2j+1)!} (\nabla)^{2j-1} ((\nabla S \cdot S)^{\rho} B_{\mu\nu\rho} S^{\mu\nu}) (S \cdot S)^{2j-1}$$

current multipoles

$$+ \sum_{j=1}^{\infty} \frac{c_{3,j}}{(2j+1)!} (\nabla)^{2j-1} ((\nabla S)^{\mu} (\nabla S)^{\nu} (S \cdot S)^{\rho\sigma}) Q_{\mu\nu\rho\sigma} (S \cdot S)^{2j-2}$$

stress multipoles

[Bianchi, Gambino, Pani, Riccioni '25] [Campanella, Riccioni '26]

3 indep. projections of Riemann

$$E_{ij} = R_{i0j0},$$

$$B_{ijk} = R_{ijk0},$$

$$Q_{ijkl} = R_{ijkl}$$

DIVERSION: HIGHER-DIMENSIONS

Higher-dim. spinning worldline

$$\begin{aligned}\mathcal{L}_R \sim & \sum_{j=1}^{\infty} \frac{c_{1,j}}{(2j)!} (\nabla)^{2j-2} (E_{\mu\nu} (S \cdot S)^{\mu\nu}) (S \cdot S)^{2j-2} \\ & + \sum_{j=1}^{\infty} \frac{c_{2,j}}{(2j+1)!} (\nabla)^{2j-1} ((\nabla S \cdot S)^{\rho} B_{\mu\nu\rho} S^{\mu\nu}) (S \cdot S)^{2j-1} \\ & + \sum_{j=1}^{\infty} \frac{c_{3,j}}{(2j+1)!} (\nabla)^{2j-1} ((\nabla S)^{\mu} (\nabla S)^{\nu} (S \cdot S)^{\rho\sigma}) Q_{\mu\nu\rho\sigma} (S \cdot S)^{2j-2}\end{aligned}$$

recovering Kerr

$$C_{n,1}^{MP} - 4C_{n,3}^{MP} \xrightarrow{D=4} 1, \quad 2C_{n,2}^{MP} \xrightarrow{D=4} 1$$

Myers-Perry multipoles

[Bianchi, Gambino, Pani, Riccioni '25]

$$C_{n,1}^{MP} = \frac{n+1+\epsilon}{2n+1} C_{n,\epsilon}$$

$$C_{n,2}^{MP} = \frac{1+\epsilon}{2} C_{n,\epsilon}$$

$$C_{n,3}^{MP} = -\frac{n}{4(2n+1)} C_{n,\epsilon}$$

$$C_{n,\epsilon} = 2(-1)^n (1+\epsilon)^{2n} \binom{n+1+\epsilon}{1+\epsilon} \binom{2n+2+2\epsilon}{1+2\epsilon}_{D=4-2\epsilon}^{-1}$$

DIVERSION: HIGHER-DIMENSIONS

Higher-dim. spinning QFTs: $D=5$ [LC, Surubaru '26]

massive little group $SO(4) \simeq SU(2) \times SU(2)$

totally symm. states $\square \square \cdots \square \sim \varepsilon^{\mu_1} \cdots \varepsilon^{\mu_s}$

(anti-)self-dual states $\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \cdots \begin{smallmatrix} \square \\ \square \end{smallmatrix} \sim \zeta_{\pm}^{\mu_1 \nu_1} \cdots \zeta_{\pm}^{\mu_s \nu_s}$ e.g. spin-1 antisymm. tensor
 $B_{\pm}^{\mu\nu} = B^{\mu\nu} \pm i\epsilon^{\mu\nu\rho}{}_{\sigma\tau} \partial_{\rho} B^{\sigma\tau}$

mixed-(A)SD states $\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \cdots \begin{smallmatrix} \square & \square \\ \square & \end{smallmatrix} \cdots \square \sim \zeta_{\pm}^{\mu_1 \nu_1} \cdots \zeta_{\pm}^{\mu_n \nu_n} \varepsilon^{\mu_{n+1}} \cdots \varepsilon^{\mu_s}$

duality preserving scattering

$$\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \cdots \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \cdots \square \quad + \quad \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \cdots \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \cdots \square \quad = \quad \# \text{ d.o.f } \mathcal{L}_R(S^{\mu\nu})$$

duality flipping scattering

$$\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \cdots \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \cdots \square \quad + \quad \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \cdots \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \cdots \square \quad - \quad = \quad \# \text{ d.o.f } \mathcal{L}_R(S^{\mu\nu}, K^{\mu})$$

extra cl. d.o.f. needed on WL

DIVERSION: HIGHER-DIMENSIONS

Higher-dim. spinning QFTs: $D=5$ [LC, Surubaru '26]

duality preserving scattering

$$+ \quad \text{[boxes]} \quad + \quad \text{[boxes]} = \# \text{ d.o.f } \mathcal{L}_R(S^{\mu\nu})$$

duality flipping scattering

$$+ \quad \text{[boxes]} \quad - \quad \text{[boxes]} = \# \text{ d.o.f } \mathcal{L}_R(S^{\mu\nu}, K^{\mu})$$

extra cl. d.o.f. needed on WL

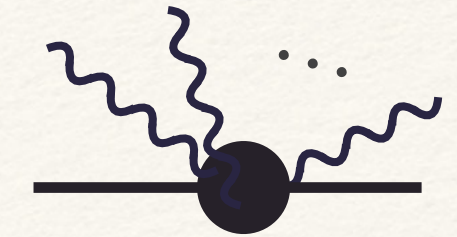
choice of states either: washes out in classical limit **or** introduces extra d.o.f

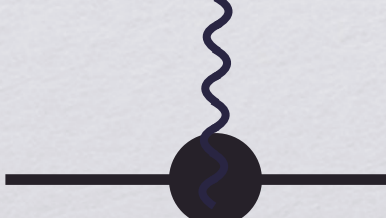
Myers-Perry amplitudes and EFTs?

can **match** against $C_{n,i=1,2,3}^{MP}$ **but underlying theories not minimal even for low spin!**

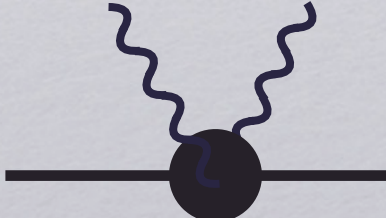
RETURN TO KERR: HIGHER ORDERS?

n-PM impulse — need coupling of n - gravitons to massive particle

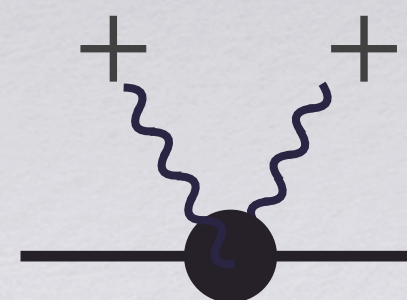


1-PM block  all orders in S

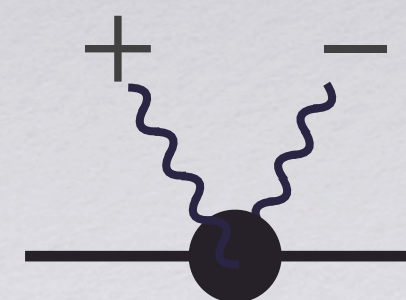
- ❖ $\mathcal{N} = 2s$ SUSY WL
- ❖ scatt. amps $\mathcal{M}(1^{(s)}, 2^{(s)}, 3^{(-2)}) = (p_1 \cdot \epsilon_3^{-1})^2 \langle \mathbf{12} \rangle^{2s}$
- ❖ massive higher-spin QFTs

2-PM block 

- ❖ $\mathcal{N} = 2$ SUSY WL up to $S^{n \leq 2s}$
- ❖ minimal spin-s th. up to $S^{n \leq 4}$
- ❖ scattering amps: using recursion
- ❖ massive higher-spin QFTs - constrain $\mathcal{L}_{R^2}$

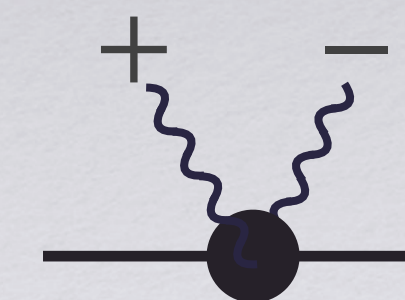


all orders in S



spurious poles
beyond $S^{n \geq 4}$

candidate amp. fixed
HS constraints
+ cl. constraints



all orders in S

HIGHER-ORDER KERR



- ❖ $\mathcal{L}_{R^2}$ captures the $\mathcal{O}(G^2)$ dynamics of linearised plane-waves off Kerr BH
- ❖ Teukolsky solutions contain it all

Laurent series in G , all-order in S , both helicity sectors

- ❖ how to read off $\mathcal{O}(G^2)$?

KERR PERTURBATIONS @ G^2

Same-helicity sector

$$\mathcal{M}_{cl} = \mathcal{M}_{Schw.} e^{-(k_1+k_2)\cdot a}$$

❖ unambiguous matching cl. amp.

❖ matches classical limit of 3pt recursion  $+ \dots = \mathcal{M}_0 \langle \mathbf{12} \rangle^{2s}$

❖ What are the worldline operators @ $\mathcal{O}(R^2)$? $\mathcal{L}_{R^{++}} = \frac{2^5}{m^2} (E_{S\mu}^+ O_{++}^{\mu\nu} E_{S\nu}^+)$

KERR PERTURBATIONS @ G²

Helicity-reversing amplitude $\mathcal{M}_{cl} = \mathcal{M}_{Schw.} e^{-(k_1+k_2)\cdot a}$

❖ unambiguous matching cl. amp.

❖ matches classical limit of 3pt recursion  + ... = $\mathcal{M}_0 \langle \mathbf{12} \rangle^{2s}$

❖ What are the worldline operators @ $\mathcal{O}(R^2)$? $\mathcal{L}_{R^{++}} = \frac{2^5}{m^2} (E_{S\mu}^+ O_{++}^{\mu\nu} E_{S\nu}^+)$

$$O_{++}^{\mu\nu} := \left(S^\mu S^\nu - \frac{1}{2} g^{\mu\nu} S^2 \right) \mathcal{F}_1^{++} - \frac{i}{m} S^2 \mathcal{F}_2^{++} \left(S^\nu \nabla^\mu - g^{\mu\nu} S \cdot \nabla \right) - \frac{1}{2} g^{\mu\nu} S^2 \mathcal{F}_3^{++}.$$

$$\begin{aligned} \mathcal{F}_1^{++}(d, b) &= \frac{(d-b)(e^{-b}-1)(e^d-1) + bd(e^{-b}+e^d-2)}{8b^2d^2} \\ &= \frac{1}{8} - \frac{5}{96}(b-d) + \frac{1}{192}(3b^2-4bd+3d^2) + \dots \end{aligned}$$

$$d := -\frac{i}{m} S \cdot \nabla, \quad b := \overleftarrow{\nabla} \cdot S \frac{i}{m}$$

KERR PERTURBATIONS @ G^2

Opposite-helicity sector

- ❖ **unambiguous matching** cl. amp. up to a^4 $\mathcal{M}_{cl} = \mathcal{M}_{Schw.} \sum_{n=0}^4 \frac{(x-w)^n}{n!}$
$$x = a \cdot (k_2 - k_1) \quad w = \langle 3|a|4] \frac{p \cdot (k_2 - k_1)}{\langle 3|p|4]}$$
- ❖ reproduced by minimal QFTs up to $s \leq 2$ & three point recursion
- ❖ ambiguities reading off cl. Tree-level amps from a^5
 - > near-zone / far-zone —> non-trivial $|a|/Gm$ dependencies $\sqrt{}$'s, $\psi^{(0)}$
- ❖ candidate amplitude from massive HS th. —> all order in a

KERR PERTURBATIONS @ G²

Opposite-helicity sector - candidate amp [LC, Chiodaroli, Johansson, Pichini, Ochirov, Skvortsov '23]

- ❖ candidate amplitude from massive HS th. $\rightarrow$ all order in a

$$\mathcal{M}_{HS} = \mathcal{M}_0 \left\{ e^x \cosh z - w e^x \sinh c z + \frac{1}{2}(w^2 - z^2)(E + 2(x - w)\tilde{E}) \right\} - \frac{(p \cdot \chi)^4}{(p \cdot q_\perp)^4} \frac{(w^2 - z^2)^2}{2} \left(\frac{\partial \tilde{E}}{\partial x} + \eta \frac{\partial \tilde{E}}{\partial z} \right) + \alpha C_\alpha$$

entire fns in spin

$$E(x, y, z) := \frac{e^y - e^x \cosh z + (x - y)e^x \sinh c z}{(x - y)^2 - z^2} + (y \rightarrow -y)$$

$$\tilde{E}(x, y, z) := \frac{1}{2y} \frac{e^y - e^x \cosh z + (x - y)e^x \sinh c z}{(x - y)^2 - z^2} + (y \rightarrow -y)$$

$$x := a \cdot q_\perp \quad y := a \cdot q$$

$$z := |a| v \cdot q_\perp$$

$$q = k_1 + k_2$$

$$q_\perp = k_2 - k_1$$

- ❖ can compute 2PM (conservative) observables: eg impulse

decomposes: parallel to BH initial momenta $\Delta p_\parallel^\mu$ + perpendicular $\Delta p_{e_\perp}^\mu, \Delta p_q^\mu$

[Bohnenblust, LC, Johansson, '24]

KERR PERTURBATIONS @ G²

2PM impulse [Bohnenblust, LC, Johansson, Pichini '24]

$$i \int d^D \mu e^{iq \cdot b} \left[q^\mu \text{ (triangle diagram) } - \int d\Phi(l_1) d\Phi(l_2) (l_1 - \bar{p}_1)^\mu \text{ (box diagram) } \right]$$

decomposes: parallel to BH initial momenta $\Delta p_{\parallel}^\mu$ + perpendicular $\Delta p_{e_{\perp}}^\mu, \Delta p_q^\mu$

$\Delta p_{\parallel}^\mu, \Delta p_{e_{\perp}}^\mu$ depend only on box coefficients $\rightarrow$ provide closed forms all-orders in spin

Δp_q^μ depend on triangle coeffs $\rightarrow$ provide results up to a^{11} + closed forms for two terms

CONCLUSIONS

- ❖ Kerr EFTs are fixed by principles (*SUSY, minimal coupling*) at low-order in G / low-order in spin
- ❖ Beyond $D = 4$, beyond a^4 , beyond low G :
lack fundamental principles to constrain the EFT
- ❖ Massive HS theories + constraints: candidate Compton & 2PM observables
- ❖ Higher-point Comptons needed for 3PM and 2PM waveform